Credit Card Fraud Detection

Objective:

- ➢ Understand the Dataset & cleanup (if required).
- ➢ Build Best fit Classification models to detect the fraud w.r.t a single & multiple feature.
- ➢ Also evaluate the models & compare their respective scores like Accuracy, precision, recall, F1-score, and confusion matrix are used to evaluate the models.

Introduction

This dataset consists of credit card transactions in the western United States. It includes information about each transaction including customer details, the merchant and category of purchase, and whether or not the transaction was a fraud.

Dataset:

- ➢ **Source:** https://www.kaggle.com/datasets/neharoychoudhury/credit-card-fraud-data/data?select=fraud_data.csv
- ➢ **Size:** 14446 records
- ➢ **Features:** Transaction Time, Purchase Category, Purchase City, Purchase State, Purchase City Locations, City Population, Amount, BirthYear, Occupation, Merchant Name, Merchant Location
- ➢ **Target Variable:** is_fraud (Whether Transaction is Fraud (1) or Not (0))

Data Description:

Column Name	Column Description
transdatetrans_time	Transaction DateTime
merchant	Merchant Name
category	Category of Merchant
amt	Amount of Transaction
city	City of Credit Card Holder
state	State of Credit Card Holder
lat	Latitude Location of Purchase
long	Longitude Location of Purchase

city_pop	Credit Card Holder's City Population
job	Job of Credit Card Holder
dob	Date of Birth of Credit Card Holder
trans_num	Transaction Number
merch_lat	Latitude Location of Merchant
merch_long	Longitude Location of Merchant
is_fraud	Whether Transaction is Fraud (1) or Not (0)

Now that data is in correct format, lets see how likely a transaction is fraudulent with respect to different features and their categories.

EDA And Features Extraction:

- [x] Transaction Year
- [x] Transaction Month
- [x] Transaction Day
- [x] Transaction Hour
- [x] Purchase Category
- [x] Purchase City
- [x] Purchase State
- [ ] Purchase City Locations
- [ ] City Population
- [ ] Amount
- [ ] BirthYear
- [ ] Occupation
- [ ] Merchant Name
- [ ] Merchant Location

Data Exploration and Visualization:

1. Purchase Category vs. Fraud:

- **Purpose:** To understand how different purchase categories relate to fraud occurrences.
- **Method:** The data is grouped by purchase categories, and the fraud counts are calculated for each category.
- **Visualization:** A bar chart is used to display the fraud counts across different purchase categories.

- **Observation:**
    i. 'purchase_category' - useful feature - since there is a difference between the fraud percentages.

2. Transaction Year vs Fraud:

- **Purpose:** To Find the relation between Transaction Year and Fraud.
- **Method:** The data is grouped by Transaction Year, and the fraud counts calculated for each category.
- **Visualization:** A bar chart is used to display the fraud counts across each year transaction
- **Observations:**
    i. There is no big difference in fraud percentage for two years
    ii. We might as well drop this column - as it is not giving much of the distinction

3. Purchase City vs. Fraud:

- **Purpose:** To analyze the distribution of fraud across different cities.
- **Method:** The data is grouped by cities, and the fraud counts are calculated.
- **Visualization:** A Plotly bar chart shows the top 10 and bottom 10 cities with the highest fraud counts. Additionally, a density map visualizes the geographical distribution of fraud.
- **Observations:**
    i. Fraud is concentrated in specific cities, which can be crucial for targeted fraud prevention measures.
    ii. We will drop the (city_lat, city_long) columns as city itself represent it

4. City Population vs. Fraud:

- **Purpose:** To investigate the relationship between city population and fraud occurrences.
- **Method:** The data is grouped by city population ranges, and the fraud counts are calculated.
- **Visualization:** A histogram or bar chart displays the distribution of fraud across different population ranges.
- **Observation:**
    i. The distribution of population of cities shows concentration of city population within half million.
    ii. Majority (Almost all) frauds are occurred for city population within 49900

iii. I'm not sure how this will help in model training/prediction - but we will take this column

5. Purchase State vs. Fraud:

- **Purpose:** To explore fraud patterns across different states.
- **Method:** The data is grouped by states, and the fraud counts are calculated.
- **Visualization:** A bar chart shows the fraud counts across states.
- **Observations:**
    i. Fraud percentage varies across states (9-16%) - However, this provides less distinction btw states w.r.t. fraud percentage
    ii. Therefore we will drop this column as well

6. Transaction Amount vs. Fraud:

- **Purpose:** To study the distribution of transaction amounts in relation to fraud.
- **Method:** The data is grouped by transaction amount ranges, and the fraud counts are calculated.
- **Visualization:** A histogram displays the distribution of transaction amounts for both fraudulent and non-fraudulent transactions.
- **Observations:**
    i. Fraudulent transactions typically fall between 0 and 1000, indicating that transaction amount is a significant feature.

7. Age vs. Fraud:

- **Purpose:** To examine the age distribution of fraud victims.
- **Method:** The data is grouped by age ranges, and the fraud counts are calculated.
- **Visualization:** A bar chart or histogram displays the age distribution of fraud victims**.**
- **Observations:**
    i. Fraud occurs across age groups from 20 to 90, with the highest concentration in the 50-60 age range.

8. Occupation vs. Fraud:

- **Purpose:** To analyze fraud occurrences across different occupations.
- **Method:** The data is grouped by occupations, and the fraud counts are calculated.
- **Visualization:** A Plotly bar chart shows the top 10 and bottom 10 occupations with the highest fraud counts.
- **Observations:**
    i. Certain occupations show higher fraud rates, making this feature useful for the model.

9. Merchant Name vs. Fraud:

- **Purpose:** To investigate fraud patterns across different merchant names.
- **Method:** The data is grouped by merchant names, and the fraud counts are calculated.
- **Visualization:** A Plotly bar chart shows the top 10 and bottom 10 merchants with the highest fraud counts.
- **Observations:**
    i. Unlike the city coordinates the merchant coordinates are not unique - a single merchant can have multiple locations - therefore in this case we will only consider the 'merchant_name' and drop the merchant coordinates.

Data Preprocessing:

1. Label Encoding:

- To convert categorical features into numerical values for model training.
- LabelEncoder is used to encode ‘purchase_category’, ‘occupation’, and ‘merchant_name’.
- This step ensures that the categorical data can be used effectively by machine learning algorithms.

2. Data Splitting:

- To split the dataset into training and testing sets while maintaining class proportions.
- Stratified sampling is used to split the data into an 80/20 ratio.
- This ensures that both the training and testing sets have a similar distribution of fraud and non-fraud cases.

Model Selection and Training:

1. Model Evaluation:

- To train and evaluate multiple machine learning models.
- Models such as RandomForestClassifier, DecisionTreeClassifier, KNeighborsClassifier, GaussianNB, and Perceptron are trained and evaluated.
- Metrics: Accuracy, precision, recall, F1-score, and confusion matrix are used to evaluate the models.
- This step helps in identifying the best performing model based on various evaluation metrics.

2. **Hyperparameter Tuning:**
    - To optimize the performance of the best performing model.
    - GridSearchCV is used to perform hyperparameter tuning for the RandomForestClassifier.
    - This step improves the model's performance by finding the best combination of hyperparameters.
3. **Confusion Matrix Visualization:**
    - To visualize the performance of the best performing model.
    - A heatmap is generated to display the confusion matrix.
    - This visualization helps in understanding the model's performance in terms of true positives, true negatives, false positives, and false negatives.
4. **Model Saving:**
    - To save the best performing model for future use.
    - The model is saved using joblib.
    - This allows the model to be easily loaded and used for future predictions without retraining.

Summary:

The project provides a comprehensive approach to credit card fraud detection, involving detailed data exploration, visualization, preprocessing, model training, evaluation, and optimization. The best performing model, a hyperparameter-tuned RandomForestClassifier, is saved for future predictions. This approach ensures that the model is well-equipped to detect fraudulent transactions effectively.

Use cases Regression and Classification:

Regression:

- ➢ **Purpose:** To predict a continuous numeric value based on input features.
- ➢ **Method:** The model is trained on historical data where the target variable is continuous. It minimizes the difference between predicted and actual values using techniques like linear regression, decision trees, or neural networks.
- ➢ **Visualization:** Scatter plots, line charts, and residual plots are often used to visualize regression predictions and errors.

➢ **Use Cases:**

- House Price Prediction: Estimating the price of a house based on features like size, location, and number of bedrooms.
- Sales Forecasting: Predicting future sales figures based on historical sales data.
- Temperature Prediction: Estimating future temperatures based on weather patterns and historical data.

Classification:

➢ **Purpose:** To predict a categorical label based on input features.
➢ **Method:** The model is trained on historical data where the target variable is categorical. It classifies data points into predefined categories using algorithms like logistic regression, decision trees, or support vector machines.
➢ **Visualization:** Bar charts, confusion matrices, and ROC curves are commonly used to evaluate classification performance.
➢ **Use Cases:**

- Spam Detection: Classifying emails as spam or not spam.
- Customer Segmentation: Categorizing customers into different groups based on purchasing behavior.
- Fraud Detection: Identifying fraudulent transactions based on patterns in the data.